

Quantitative Topology and Geodesics

Isabel Beach
University of Toronto

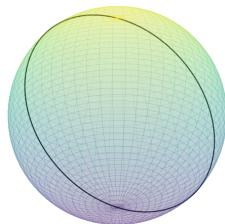
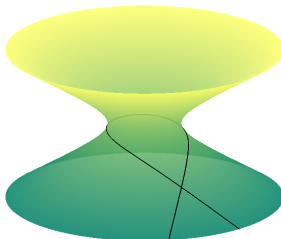
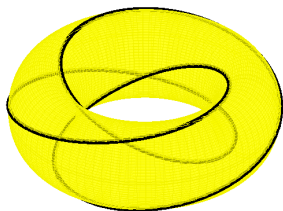
September 10 2025

Geodesics

Definition

A geodesic is a curve γ satisfying the equation

$$\frac{D}{dt} \left(\frac{d\gamma}{dt} \right) = 0.$$



Quantitative Topology

Question

Can we make quantitative versions of algebraic existence theorems for geometric objects?

We want to create quantitative versions of algebraic existence theorems for geodesics.

Example: Existence Theorem

There exists a geodesic....

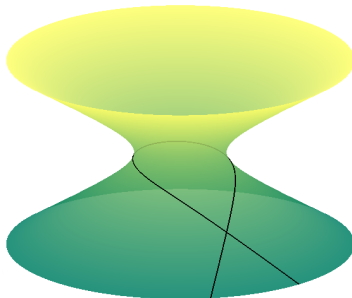
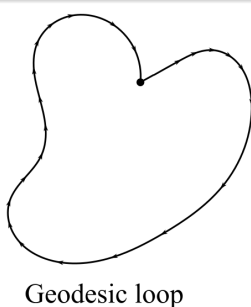
Example: Quantitative Theorem

There exists a geodesic and it has length bounded by...

Geodesic Segments and Loops

Definition

A geodesic loop is a geodesic that starts and ends at the same point.



Geodesic Segments and Loops

Serre 1951 [19]

On a closed Riemannian manifold M , there are infinitely many geodesics connecting every pair of points p, q .

Nabutovsky–Rotman 2013 [15]

On a closed Riemannian surface M , between any points p, q there are k geodesics **of length at most** $22k \operatorname{diam} M$, or $20k \operatorname{diam} M$ if $p = q$, for all $k > 0$.

Cheng 2022 [5]

On a closed Riemannian surface M , between any points p, q there are k geodesics **of length at most** $8k \operatorname{diam} M$, or $6k \operatorname{diam} M$ if $p = q$, for all $k > 0$.

Geodesic Segments and Loops

If $p = q$, we get infinitely many geodesic loops. However, these k loops might just be the same periodic geodesic iterated k times. How can we obtain truly distinct loops?

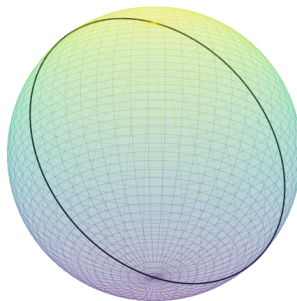
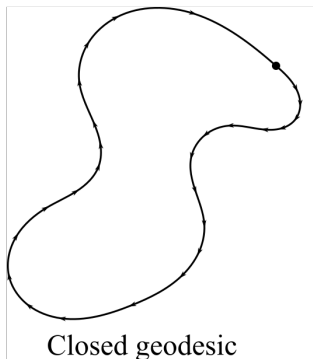
B. 2024

Given any point p in a Riemannian 2-sphere M , there are two distinct **simple** geodesic loops based at p **of length at most** $7 \operatorname{diam}(M)$ and $14 \operatorname{diam}(M)$.

Closed Geodesics

Definition

A closed geodesic is a geodesic loop whose endpoints meet smoothly.



Closed Geodesics

Fet–Lyusternik 1951 [13]

Any compact Riemannian manifold M admits a closed geodesic.

Question (Gromov)

Is there bound for the length of a shortest closed geodesic on a Riemannian manifold M in terms of $\sqrt[n]{\text{vol}(M)}$?

When M is a compact surface but not a 2-sphere, answers in various cases were given by P. Pu, C. Loewner, M. Gromov, J. Hebda, Y. Burago, V. Zalgaller, and others.

Closed Geodesics

Birkhoff 1917 [3]

Every Riemannian 2-sphere contains at least one closed geodesic.

Every Riemannian 2-sphere of area A contains at least one closed geodesic of length at most L , where...

Croke 1988 [6]

$$L \leq 31\sqrt{A}$$

Sabourau 2004 [17], Rotman–Nabutovsky 2002 [14]

$$L \leq 8\sqrt{A}$$

Rotman 2006 [16]

$$L \leq 4\sqrt{2A}$$

Closed Geodesics

Thorbergsson 1978 [20]

Any complete Riemannian surface M with finite area A that **is not** homeomorphic to $\mathbb{R}^2$ or $\mathbb{R} \times S^1$ admits a closed geodesic.

Bangert 1980 [1]

Any complete Riemannian surface M with finite area A that **is** homeomorphic to $\mathbb{R}^2$ or $\mathbb{R} \times S^1$ admits a closed geodesic.

Thus, any complete Riemannian surface M with finite area A admits a closed geodesic.

Closed Geodesics

Croke 1981 [6]

Any complete Riemannian surface M with finite area A admits a closed geodesic **of length at most** $31\sqrt{A}$.

B.-Rotman 2020 [2]

Any complete Riemannian surface M with finite area A admits a closed geodesic **of length at most** $4\sqrt{2A}$.

Closed Geodesics

Lyusternik–Schnirelmann 1930

Every Riemannian 2-sphere contains three simple closed geodesics.

Liokumovich–Nabutovsky–Rotman 2017 [12]

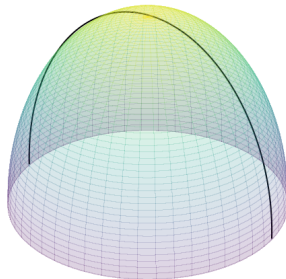
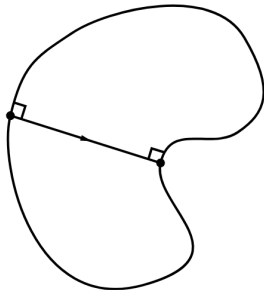
Every Riemannian 2-sphere contains three simple closed geodesics **of respective lengths at most** 5 diam M , 10 diam M , and 20 diam M .

Orthogonal Geodesic Chords

Definition

An orthogonal geodesic chord is a geodesic segment in a manifold with boundary that:

- 1 meets the boundary orthogonally at its endpoints.
- 2 does not intersect the boundary at any other point.



Background: Brake Orbits

Consider the Hamiltonian system associated with

$$H(p, q) = \frac{1}{2} p^T A(q) p + V(q)$$
$$A : \mathbb{R}^n \rightarrow \mathbb{R}^n \times \mathbb{R}^n, V : \mathbb{R}^n \rightarrow \mathbb{R}$$

Using the metric $g = \frac{1}{2} A(q)$, this Hamiltonian system is equivalent to

$$\frac{D}{dt} \dot{q} + \nabla V(q) = 0$$

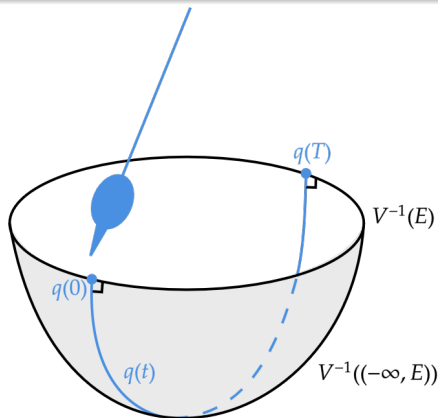
Each solution q has energy E defined by

$$E = \frac{1}{2} \|\dot{q}\|^2 + V(q)$$

Background: Brake Orbits

Definition

A **brake orbit** with energy E is a solution curve $t \mapsto (p(t), q(t))$ with $p(0) = p(T) = 0$ for some T .



Background: Brake Orbits

Assume that E is a regular value of V and $V^{-1}((-\infty, E])$ is homeomorphic to an n -disk.

H. Seifert (1949) [18]

There is at least one brake orbit with energy E .

Seifert's Conjecture

There are at least n (distinct) brake orbits with energy E .

Giambò–Giannoni–Piccione 2005 [7]

There is a $\delta > 0$ such that an **orthogonal geodesic chord** on $(V^{-1}((-\infty, E - \delta]), g_E)$ can be extended to a unique **brake orbit** on $V^{-1}((-\infty, E])$.

Orthogonal Geodesic Chords

Bos 1963 [4]

A convex n -disk admits at least n orthogonal geodesic chords.

Gluck–Ziller 1984 [8]

A manifold with convex boundary admits at least one orthogonal geodesic chord.

Orthogonal Geodesic Chords

Scott–Hass 1994 [9], Ko 2023 [10]

A 2-disk M with convex boundary admits two simple orthogonal geodesic chords.

B. 2025

A 2-disk M with convex boundary admits two simple orthogonal geodesic chords **of length at most**

$$L(\partial M) + 4 \operatorname{diam} M$$

and

$$2L(\partial M) + 2 \operatorname{diam} M + 686\sqrt{A(M)}.$$

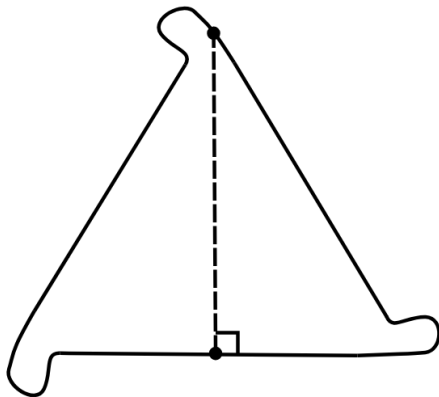
Orthogonal Geodesic Chords

Question

What if ∂D is not convex?

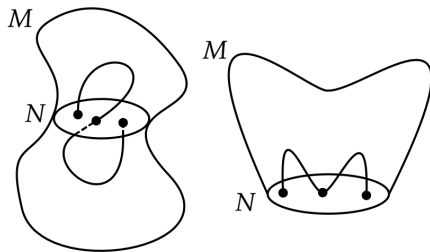
By Bos, we know that not every non-convex disk admits an orthogonal geodesic chord.

However, we still get chords orthogonal at one end.



Geodesic Chords Orthogonal to Submanifolds

We can also consider geodesic segments that meet a given submanifold orthogonally at their endpoints.



Why We Care

These curves generalize our previous results. In the case that N is a point, these curves are geodesic loops. In the case that $N = \partial M$, these curves are orthogonal geodesic chords (in the usual sense).

Geodesic Chords Orthogonal to Submanifolds

B., Contreras Peruyero, Griffin, Kerr, Rotman, Searle
2025

Let M be a compact n -dimensional manifold, and $N \subset M$ a closed submanifold.

Suppose M satisfies the “effective loop shortening property restricted to N with parameter c ”^{*}.

Then there exists a geodesic chord in M that is orthogonal to N at its endpoints and has length at most $2c(2n + 1)$.

^{*}Roughly, any long-ish loop based a point in N can be shortened a substantial amount through loops of length $\leq c$.

Example Applications

Liokumovich, Nabutovsky, Rotman 2015 [11]

Any 2-disk Ω can be contracted to a point through loops of length at most $2L(\partial\Omega) + 686\sqrt{\text{Area}(\Omega)} + 2 \text{diam } \Omega$.

Therefore the effective loop shortening property holds on 2-spheres!

Then for any submanifold $S^2 \simeq N$ of a n -dimensional manifold M , there is a geodesic chord in M orthogonal to N of length

$$\leq \left(4 \text{diam } M + 96 \text{diam } N + 8232\sqrt{\text{Area}(N)}\right) (2n + 1).$$

Strategy

B. 2025

A 2-disk M with convex boundary admits two simple orthogonal geodesic chords **of length at most**

$$L(\partial M) + 4 \operatorname{diam} M$$

and

$$2L(\partial M) + 2 \operatorname{diam} M + 686\sqrt{A(M)}.$$

How do we prove these types of results?

Strategy

Consider a Riemannian 2-disk M .

Definition

ΩM is the set of piecewise differentiable paths in M with endpoints on ∂M that have no transverse self-intersections.

$\Omega_0 M$ is the set of constant paths on ∂M .

How to find a pair of simple orthogonal geodesic chords:

- Define a length shortening flow on ΩM .
- Find non-trivial classes $h_i \in H_i(\Omega M, \Omega_0 M)$ for $i = 1, 2$ such that $\omega \cap h_2 = h_1$ for some ω .
- Find maps ϕ_i that represent h_i . Let ℓ_i be the maximum length of a curve in the image of ϕ_i .

Strategy

- Apply the shortening flow to the curves in the image of ϕ_i .
- A subsequence of the longest curves will converge to a simple orthogonal geodesic chord γ_i of length at most ℓ_i .
- If $\gamma_1 = \gamma_2$, then we can use a Lyusternik–Schnirelmann argument to prove there are infinitely many distinct orthogonal geodesic chords of the same length.

We then obtain a pair of distinct simple orthogonal geodesic chords of lengths ℓ_1 and ℓ_2 .

Step One: A Length Shortening Flow

Hass-Scott 1994 [9]

There is a curve shortening flow with the following important properties.

- 1 This flow does not create transverse self-intersections (so it really is a flow on ΩM).
- 2 Curves converge under this flow (up to a subsequence) to simple orthogonal geodesic chords or points.
- 3 When applying this flow to a k -parameter family of curves, the new family is homotopic to the original family.

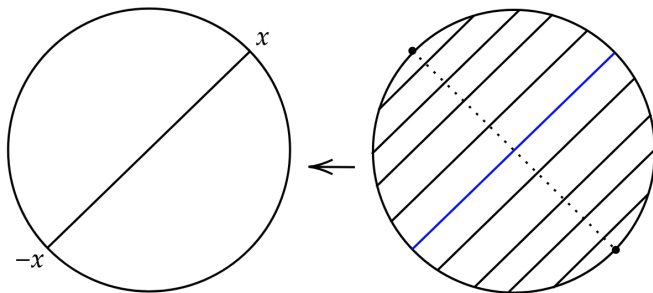
Step Two: Finding Suitable Homology Classes

Fact

For $i = 0, 1, 2$, we have $H_i(\Omega M, \Omega_0 M, \mathbb{Z}_2) \simeq \mathbb{Z}_2$.

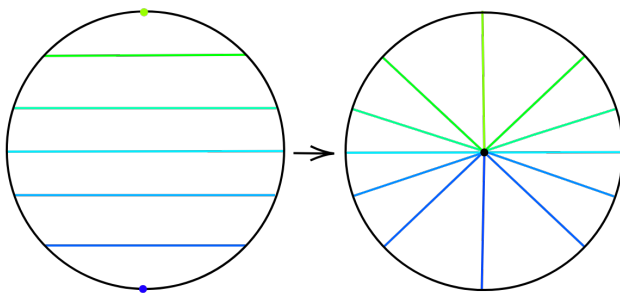
In the Euclidean disk...

- 1 The 1-dimensional class is represented by the set of all chords parallel to the equator.
- 2 The 2-dimensional class is represented by the set of all chords.



Step Two: Finding Suitable Homology Classes

We can also represent the classes using pairs of radial geodesics by mapping a chord to the two radii connecting its endpoints to the centre of the disk.



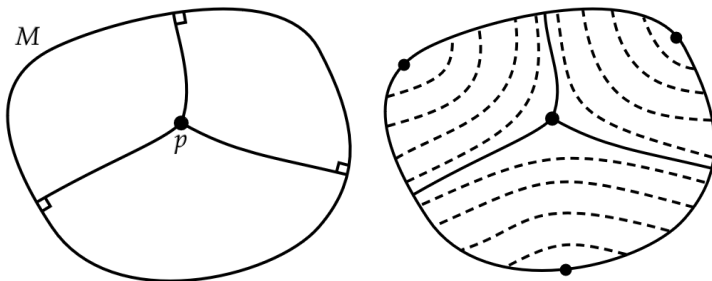
Therefore we want to find an analog of “radial” curves in M .

Step Three: Finding a Sweepout

Pick a point $p \in M$ at maximal distance from ∂M to be our centre.

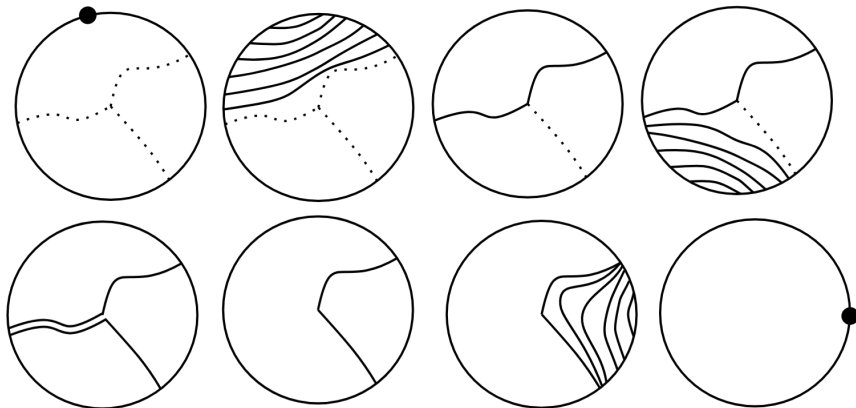
Using Berger's Lemma, divide M into convex regions bounded by pairs of minimizing geodesics emanating from p . These geodesics have length at most $\text{diam } M$.

Then try to shorten each pair of geodesics to a point curve. Each homotopy will stay in its own convex region.



Step Three: Finding a Sweepout

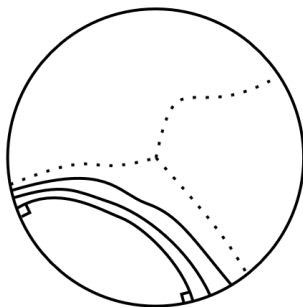
If we succeed, we combine these homotopies to sweep out M .



Step Three: Finding a Sweepout

However, we might get stuck on a simple orthogonal geodesic chord of length at most $2 \operatorname{diam} M$ instead of shortening to a point.

To find a second simple chord we still need to contract our regions.



Step Three: Finding a Sweepout

Liokumovich–Nabutovsky–Rotman 2015 [11]

Any disk M can be contracted through a homotopy of loops on ∂M of length at most $2L(\partial M) + 2 \operatorname{diam} M + 686\sqrt{A(M)}$.

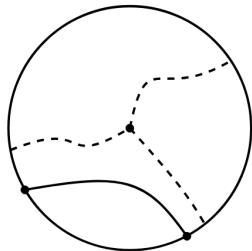
We can modify this homotopy to pass through simple, non-mutually intersecting curves.

This lets us finish contracting our regions.

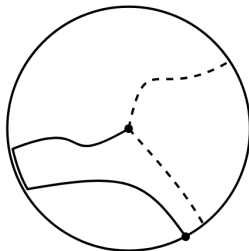
Step Three: Finding a Sweepout

At this point, we have devised a way to contract all the regions through curves of length at most ℓ .

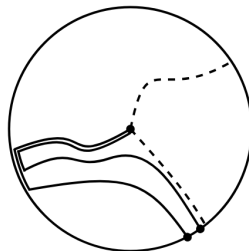
Curves in our sweepout:
Length $\leq \ell$



Curves in our radial sweepout:
Length $\leq \ell + L(\partial M) + \text{diam } M$



Curves in our representatives:
Length $\leq 2(\ell + L(\partial M) + \text{diam } M)$



Now we have representatives of our two homology class.

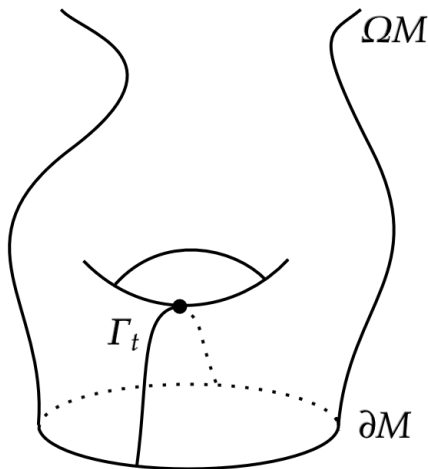
Step Four: Pulling Tight

Apply the free boundary flow to the curves in the images of these representatives.

Since these maps represent non-trivial homology classes, they cannot contract to a point.

This is because they “get stuck” on a critical point of the length functional (by Morse theory).

These critical points are simple orthogonal geodesic chords.



Future Work

Multiple orthogonal geodesic chords:

Bos 1963 [4]

There are at least n distinct orthogonal geodesic chords in a convex n disk.

Quantitative Version

There are at least n distinct orthogonal geodesic chords in a convex n disk **of length at most** ...

“Higher dimensional” orthogonal geodesic chords:

Minimal Surfaces

Let M be a manifold with boundary. There exists a free boundary minimal submanifold in M **of volume at most** ...

Thank you!

Section 1

References

- [1] V. Bangert. Closed geodesics on complete surfaces. *Mathematische Annalen*, 251(1):83–96, 1980.
- [2] I. Beach and R. Rotman. The length of the shortest closed geodesic on a surface of finite area. *Proceedings Of The American Mathematical Society*, 2020.
- [3] G. Birkhoff. *Dynamical Systems*. American Mathematical Society. American Mathematical Society, 1927. ISBN 9780821810095.
- [4] W. Bos. Kritische sehenen auf riemannischen elementarraumstücken. *Mathematische Annalen*, 151(5): 431–451, 1963.
- [5] H. Y. Cheng. Curvature-free linear length bounds on geodesics in closed riemannian surfaces. *Transactions of the American Mathematical Society*, 375(7):5217–5237, 2022.
- [6] C. B. Croke. Area and the length of the shortest closed geodesic. *J. Differential Geom.*, 27(1):1–21, 1988.

- [7] R. Giambó, F. Giannoni, and P. Piccione. Orthogonal geodesic chords, brake orbits and homoclinic orbits in riemannian manifolds. *Advances in Differential Equations*, 10(8):931–960, 2005.
- [8] H. Gluck and W. Ziller. Existence of periodic motions of conservative systems. In E. Bombieri, editor, *Seminar On Minimal Submanifolds*, volume 103, pages 65–98. Princeton University Press, Princeton, 1984.
- [9] J. Hass and P. Scott. Shortening curves on surfaces. *Topology*, 33(1):25 – 43, 1994.
- [10] D. Ko. Existence and morse index of two free boundary embedded geodesics on riemannian 2-disks with convex boundary, 2023. URL <https://arxiv.org/abs/2309.09896>.
- [11] Y. Liokumovich, A. Nabutovsky, and R. Rotman. Contracting the boundary of a riemannian 2-disc. *Geometric and Functional Analysis*, 25(5):1543 – 1574, 2015.
- [12] Y. Liokumovich, A. Nabutovsky, and R. Rotman. Lengths of

- three simple periodic geodesics on a Riemannian 2-sphere. *Mathematische Annalen*, 367:831–855, 2017.
- [13] L. Lyusternik and A. Fet. Variational problems on closed manifolds. *Doklady Akademii Nauk SSSR*, 81:17–18, 1951.
- [14] A. Nabutovsky and R. Rotman. The length of a shortest closed geodesic on a 2-dimensional sphere. *IMRN*, 2002(39): 2121–2129, 2002.
- [15] A. Nabutovsky and R. Rotman. Linear bounds for lengths of geodesic segments on riemannian 2-spheres. *Journal of Topology and Analysis*, 05(04):409–438, 2013.
- [16] R. Rotman. The length of a shortest closed geodesic and the area of a 2-dimensional sphere. *Proceedings of the American Mathematical Society*, 134(10):3041–3047, 2006.
- [17] S. Sabourau. Filling radius and short closed geodesics of the 2-sphere. *Bulletin de la Société Mathématique de France*, 132(1):105–136, 2004.
- [18] H. Seifert. Periodische bewegungen mechanischer systeme. *Mathematische Zeitschrift*, 51:197–216, 1949.

- [19] J. Serre. Homologie singulière des espaces fibrés. Applications. *Annals of Mathematics*, 54:425–505, 1951.
- [20] G. Thorbergsson. Closed geodesics on non-compact Riemannian manifolds. *Mathematische Zeitschrift*, 159(3): 249–258, 1978.